

# Sean Bowman

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## SUMMARY

Responsible engineer in Blue Origin's New Glenn Stage 2 Fluids group for the heat exchanger that supplies tank pressurization in flight, now in qualification testing. Previously Director of Propulsion at Vaya Space, leading the engineering organization behind the Dauntless supercritical oxygen expander cycle hybrid rocket engine from design through test operations and flight qualification. Background spans hybrid engine development, thermo-fluid analysis, and engineering software architecture.

## EXPERIENCE

### Blue Origin

Merritt Island, FL

#### Fluid System Engineer III – New Glenn Stage 2 Fluids, Hydrazine

August 2026 – Present

- Responsible engineer for the heat exchanger that supplies pressurization fluid to the second-stage tanks in flight
- Oversee the heat exchanger's qualification testing

### Vaya Space

Cocoa, FL

#### Director of Propulsion

January 2026 – June 2026

*Chief Engineering Council: one of four department directors (GNC, Avionics, Structures, Propulsion) collectively holding company-wide Chief Engineer authority; set engineering standards, technical levels, and program priorities atop full Director-level departmental scope.*

- Led the propulsion department accountable for all deliverables on the Dauntless hybrid engine, a supercritical oxygen expander cycle with swirl injection, across the full system lifecycle: design, analysis, manufacturing, test operations, and flight qualification
- Directed a team of 8 engineers and developers; owned departmental schedule, personnel, and budget, and carried program management and mission assurance for the propulsion program
- Architected the 38,000+-line Python propulsion design suite with 100+ documentation files; established coding standards, onboarding, and a main/dev/feature Git workflow with mandatory reviews and quarterly releases
- Designed multi-user frontend infrastructure for concurrent engineering operation: priority-based CPU allocation, background job processing with live progress streaming, and session management with automatic resource cleanup

#### Aerospace Engineer II – Fluid Thermal Control

April 2024 – January 2026

- Took full product ownership of the regeneratively-cooled nozzle: end-to-end contour and cooling-channel design via Axisymmetric Method of Characteristics, 3D spirally-fluted geometry for LPBF additive manufacturing in GRCop-42, and CAD release
- Served as responsible engineer through nozzle fabrication and test: vendor and manufacturing oversight, DFM reviews, part inspection and assembly, control-room operations, and post-test validation against design predictions

#### Aerospace Engineer I – Fluid Thermal Control

April 2022 – April 2024

- Designed, analyzed, and tested the regenerative cooling architecture of the Dauntless nozzle: LOX coolant path design for supercritical oxygen operation and conjugate heat transfer analysis with REFPROP/CEA integration
- Originated the propulsion design suite, authoring the initial MATLAB implementation and later porting it to an object-based Python library to cut licensing cost and broaden access; expanded it from a focused regenerative-cooling toolset to the full engine design environment

### Graduate Research Assistant

Florida Institute of Technology

Spring 2021 – Fall 2021

Melbourne, FL

- Performed DPM CFD simulation of cryogenic droplet thermo-fluid mixing; achieved <5% deviation from experimental benchmarks
- Delivered validated analysis tool to leading rocket manufacturer; analysis article currently in active use

### Graduate Teaching Assistant Lead

Florida Institute of Technology

January 2019 – May 2021

Melbourne, FL

- Created course materials and lectured for AEE 3064 Fluid Mechanics Laboratory; oversaw a team of instructors conducting weekly progress meetings

## TECHNICAL SKILLS

**Programming:** Python, MATLAB, C++, JavaScript, HTML, FORTRAN, VBS, Git

**Engineering Tools:** Siemens NX, Teamcenter (PLM), Simcenter Nastran, STAR-CCM+, ANSYS Fluent, SolidWorks, Atlassian Suite (Jira, Confluence, Bitbucket), LaTeX, REFPROP, CEA, GFSSP

**Process:** Responsible Engineering & Cradle-to-Grave Hardware Ownership, Design Reviews (PDR/CDR/MRR/TRR), Cross-Functional Coordination, Vendor & Supplier Management, Test Planning, Readiness & Operations, Risk Assessment, Program Management, Mission Assurance, Design for Manufacturability & Assembly, Additive Manufacturing, Code Review, Technical Documentation

**Leadership:** Team Direction & Mentorship, Standups & 1:1s, Executive & Board Engagement, Resource & Budget Planning, Engineering Standards & Infrastructure, Investor Relations, DoD / NASA Interface

## EDUCATION

### Master of Science in Aerospace Engineering

December 2021

Florida Institute of Technology

Melbourne, FL

### Bachelor of Science in Applied Physics, Minor: Mathematics

December 2017

Stockton University, Magna Cum Laude

Galloway, NJ